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ENHANCED METHOD TO TROUBLESHOOT AIRCRAFT ELECTRIC POWER GENERATING SYSTEM FAULT EVENTS

Abstract

A fault analysis system, including: a memory device including a circular buffer, and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer. The one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device. The pre-fault snapshot data includes a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event. The fault snapshot data includes a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

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Background/Summary

BACKGROUND

[0001] The subject matter disclosed herein generally relates to fault events in an aircraft electric power generating system (EPGS), and more particularly to enhanced troubleshooting of aircraft EPGS fault events.

BRIEF DESCRIPTION

[0002] A fault analysis system is disclosed, including: a memory device including a circular buffer; and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer; wherein in the one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device, wherein: the pre-fault snapshot data includes a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data includes a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

[0003] Any one or combination of the foregoing embodiments, further including: one or more second processors configured to generate fault analysis data associated with the fault event based on processing, by the one or more second processors, the pre-fault snapshot data and the fault snapshot data.

[0004] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes a rate of change associated with at least one status signal of the status signals.

[0005] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes a pattern associated with at least one status signal of the status signals.

[0006] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.

[0007] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault event is of a different type compared to the fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault condition is of a different type compared to the fault condition.

[0008] Any one or combination of the foregoing embodiments, wherein the fault analysis system is configured to set the temporal period based on a type of the fault event.

[0009] Any one or combination of the foregoing embodiments, wherein the fault analysis system is configured to detect the occurrence of the fault event based on receiving a fault-confirmation signal.

[0010] A fault reporting system is disclosed, including: a memory device including a circular buffer; and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer; wherein in the one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device, wherein: the pre-fault snapshot data includes a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data includes a second portion of the stored status signals corresponding to the time point associated with

occurrence of the fault event.

[0011] Any one or combination of the foregoing embodiments, wherein the fault reporting system is configured to provide the pre-fault snapshot data and the fault snapshot data in association with generating fault analysis data associated with the fault event.

[0012] Any one or combination of the foregoing embodiments, wherein the fault reporting system is configured to set the temporal period based on a type of the fault event.

[0013] Any one or combination of the foregoing embodiments, wherein the fault reporting system is configured to detect the occurrence of the fault event based on receiving a fault-confirmation signal.

[0014] A method is disclosed, including: continuously capturing, by one or more processors, status signals associated with an electric power generating system of an aircraft and storing the status signals to a circular buffer; and in response to detecting a fault event associated with the electric power generating system, storing pre-fault snapshot data and fault snapshot data associated with the fault event to a memory device, wherein: the pre-fault snapshot data includes a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data includes a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

[0015] Any one or combination of the foregoing embodiments, further including: generating fault analysis data associated with the fault event based on processing, by a computing device, the pre-fault snapshot data and the fault snapshot data.

[0016] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes a rate of change associated with at least one status signal of the status signals.

[0017] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes a pattern associated with at least one status signal of the status signals.

[0018] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.

[0019] Any one or combination of the foregoing embodiments, wherein the fault analysis data includes an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault event is of a different type compared to the fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault condition is of a different type compared to the fault condition.

[0020] Any one or combination of the foregoing embodiments, further including setting the temporal period based on a type of the fault event.

[0021] Any one or combination of the foregoing embodiments, wherein detecting the occurrence of the fault event is based on receiving a fault-confirmation signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike.

[0023] FIG. 1 is a schematic illustration of a system that can incorporate various embodiments of the present disclosure.

[0024] FIG. 2 is an example timeline supportive of enhanced troubleshooting of aircraft fault events in accordance with example aspects of the present disclosure.

[0025] FIG. 3 illustrates an example flowchart of a method that supports enhanced troubleshooting of aircraft fault events in accordance with one or more embodiments of the present disclosure.

[0026] FIG. 4 illustrates an example flowchart of a method that supports enhanced troubleshooting of aircraft fault events in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0027] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

[0028] FIG. 1 is a schematic illustration of a system **100** that can incorporate various embodiments of the present disclosure. As shown in FIG. 1, the system **100** may include an aircraft **101** and one or more computing devices **103**.

[0029] The computing devices **103** may be capable of performing any combination of operations described herein. In some cases, the system **100** may include computing devices **103** internal to and integrated with the aircraft system **102** and computing devices **103** separate from the aircraft system **102**. Components of the aircraft system **102** described herein may be electrically coupled via any combination of buses (not illustrated) (e.g., power buses, data buses, avionics buses, or the like).

[0030] A computing device **103** (e.g., computing device **103-a**, computing device **103-b**) may be disposed in operable communication with components of the aircraft **101**. The system **100** and aircraft system **102** supports communication between the computing devices **103** and other devices of the system **100** via wired communication protocols, wireless communication protocols (e.g., electromagnetic (EM) signals, WiFi, Bluetooth™, ZigBee™, Ubiquiti™, 3G, 4G, LTE, and the like), and/or combinations including one or more of the foregoing.

[0031] The computing device **103** is configured to receive, store and/or transmit data. The computing device **103** includes processing components configured to analyze received signals and data described herein. The computing device **103** includes processing components configured to provide data (and/or control signals) to other components of the system **100** or aircraft system **102**. The computing device **103** includes any number of suitable components, such as processors, memory, communication devices and power sources.

[0032] The computing device **103** may include processing circuitry capable of executing instructions stored on a memory of the computing device **103** in association with performing one or more functions described herein. Some elements stored in the memory may be described as or referred to as instructions or instruction sets, and some functions of the computing device **103** may be implemented using machine learning techniques.

[0033] The aircraft **101** can include an EPGS **105**. The EPGS **105** may include an EPGS controller **110**. In some examples, logic circuitry **115** may be implemented in the EPGS controller **110**. The logic circuitry **115** may be, for example, built-in test equipment (BITE) logic. In some embodiments, the EPGS **105** may include multiple EPGS controllers **110**. It is to be understood that aspects described herein with reference to a EPGS controller **110** may be applied to other EPGS controllers **110** included in the EPGS **105**.

[0034] The EPGS controller **110** may include microcontroller **116** (or multiple microcontrollers **116**). The microcontroller **116** may correspond to one or many computer processing devices. For example, the **116** may include a silicon chip, such as a FPGA, an ASIC, any other type of IC chip, a collection of IC chips, or the like. In some aspects, the processors may include a microprocessor, CPU, a GPU, or plurality of microprocessors configured to execute the instructions sets stored in a corresponding memory (e.g., memory **117** of the microcontroller **116**). For example, upon executing the instruction sets stored in memory **117**, the microcontroller **116** may enable or perform one or more functions described herein. Some elements stored in memory **117** may be described as or referred to as instructions or instruction sets.

[0035] The memory **117** may include one or multiple computer memory devices. The memory **117** may include, for example, Random Access Memory (RAM) devices, Read Only Memory (ROM) devices, flash memory devices, magnetic disk storage media, optical storage media, solid state storage devices, core memory, buffer memory devices, combinations thereof, and the like. The

memory **117**, in some examples, may correspond to a computer readable storage media. In some aspects, the memory **117** may be internal or external to the EPGS controller **110** and/or the microcontroller **116**.

[0036] The memory **117** may be configured to store instruction sets and other data structures (e.g., depicted herein) in addition to temporarily storing data for the microcontroller **116** to execute various types of routines or functions. For example, the memory **117** may be configured to store program instructions (instruction sets) that are executable by the microcontroller **116** and provide functionality of the EPGS controller **110** described herein. The memory **117** may also be configured to store data or information that is useable or capable of being called by the instructions stored in memory **117**.

[0037] The aircraft **101** may include multiple line replaceable units (LRUs) **120** electrically coupled to the EPGS **105**. The LRUs **120** may be coupled to the EPGS **105** via one or more data buses. Non-limiting examples of the LRUs **120** include generators, contactors, controllers, switches, wiring, sensors, and the like.

[0038] The EPGS **105** may include a non-volatile memory (NVM) device **125** capable of storing data. Aspects of the present disclosure include implementing a circular buffer **130** in the NVM device **125**. The circular buffer **130** (also referred to herein as a circular queue, a cyclic buffer, or a ring buffer) is a data structure that uses a single buffer as if the buffer were connected end-to-end. The size of the circular buffer **130** may be a fixed size or may be tunable by the system **100**.

[0039] The circular buffer **130** may have a predetermined storage space. In some cases, the circular buffer **130** may be capable of continuously storing data. For example, a property of the circular buffer **130** is that when the circular buffer **130** is full and a subsequent writing of data to the circular buffer **130** is performed, the data write may overwrite the oldest data stored on the circular buffer **130**. In some aspects, the techniques described herein include setting the storage space of the circular buffer **130** based on one or more criteria.

[0040] According to one or more embodiments of the present disclosure, the system **100** supports enhanced techniques of capturing and storing information relevant to a fault event associated with the EPGS **105** of the aircraft **101**. The techniques described herein provide data (snapshot data record **135** later described herein) supportive of improving the accuracy and efficiency of field service engineering and product engineering analyses when an aircraft event is reported.

[0041] According to one or more embodiments of the present disclosure, in response to a fault event at the aircraft **101** (e.g., at a LRU **120**), the EPGS controller **110** may isolate the reason for the fault event to the most relevant LRU **120** in the aircraft system **102**. In some examples, the fault event may be due to one or more fault conditions, example aspects of which are later described herein.

[0042] In an example, circuitry in the EPGS controller **110** may provide control signals **111** to microcontroller **116** to consume. For example, the EPGS controller **110** provides, in digital form (e.g., control signals **111**), conditions of switches and measurement data (e.g., temperatures, pressures, and the like) to microcontroller **116** to consume. Based on the control signals **111**, the microcontroller **116** may produce status signals **112**.

[0043] The EPGS controller **110** (processor) may act on the status signals **112** to determine whether a fault condition is present. For example, the EPGS controller **110** (e.g., using logic circuitry **115**) may generate and output a fault-confirmation signal **114** indicating the presence of the fault condition. That is, for example, the EPGS controller **110** may generate a fault code (e.g., a fault-confirmation signal **114**) for confirming the fault event. In an example, a fault event may include a generator failure associated with the aircraft system **102**, but is not limited thereto.

[0044] Non-limiting examples of the status signals **112** include voltages, currents, frequencies, temperatures, internal controller statuses, switches, contactors, and the like. The aircraft system **102** (e.g., at EPGS controller **110**) may store, to snapshot data record **135**, fault snapshot data **140-b** (inclusive of the status signals **112**) and an associated isolated fault code to the NVM device **125**

for engineering investigation support. For example, the snapshot data record **135** and fault snapshot data **140-b** may support post-event analysis in accordance with example aspects of the present disclosure.

[0045] In accordance with one or more embodiments of the present disclosure, the EPGS **105** may be capable of providing pre-fault snapshot data **140-a** and fault snapshot data **140-b** supportive of post-event analysis (e.g., post fault event). In an example, the EPGS **105** (e.g., at EPGS controller **110**) may continuously capture the same status signals **112** (e.g., input and output signals) and store the status signals **112** to circular buffer **130**.

[0046] In the event of a detected fault, the aircraft system **102** may provide a portion of the status signals **112** (stored in the circular buffer **130**) to the snapshot data record **135** as pre-fault snapshot data **140-a**. For example, in response to a fault-confirmation signal **114** generated by the EPGS **105** (e.g., at microcontroller **116**), the EPGS **105** may further generate the pre-fault snapshot data **140-a**. In some embodiments, the EPGS **105** may provide the status signals **112** for a target temporal period (e.g., temporal period **232** later described herein with reference to FIG. 2) prior to the fault event, based on a type or classification of the fault event.

[0047] In some embodiments, the EPGS **105** may be capable of providing pre-fault snapshot data **140-a** and fault snapshot data **140-b**, along with the fault-confirmation signal **114** (or multiple fault-confirmation signals **114**) in the snapshot data record **135** associated with a given fault event. Accordingly, for example, the further inclusion of the pre-event (pre-trigger) status signals included in the pre-fault snapshot data **140-a** support enhanced post-event engineering analysis of a fault by providing additional insight into the status of the generating channel/system leading up to the fault confirmation.

[0048] For example, the EPGS **105** may provide the snapshot data record **135** to a computing device **103** (e.g., computing device **103-a**, computing device **103-b**, example aspects of which are described herein). Based on the pre-fault snapshot data **140-a** and the fault snapshot data **140-b** included in the snapshot data record **135**, the computing device **103** may generate fault analysis data **145** (also referred to herein as insight data) of factors leading up to the fault confirmation. Non-limiting examples of the fault analysis data **145** include a rate of change in voltage, a rate of change in frequency, a rate of change in temperature, and the like.

[0049] FIG. 2 is an example timeline **200** supportive of enhanced troubleshooting of aircraft EPGS fault events in accordance with example aspects of the present disclosure. The timeline **200** illustrates aspects of storing pre-fault snapshot data **140-a** and fault snapshot data **140-b** to a snapshot data record **135** (also referred to herein as a BITE fault record) in response to confirmation of a fault event. Aspects of the timeline **200** of FIG. 2 will be described herein with reference to the system **100** of FIG. 1.

[0050] According to one or more embodiments of the present disclosure, the system **100** may support fault reporting by the aircraft system **102** and fault analysis by computing device **103-a** (e.g., a central maintenance computer included in the aircraft system **102**) or computing device **103-b** (e.g., based on data provided by the aircraft system **102**).

[0051] In some examples, in response to detection of a fault event, the EPGS **105** is configured to store pre-fault snapshot data **140-a** and fault snapshot data **140-b** associated with the fault event to the memory device **125**. In some examples, the fault event may be based on a fault code (e.g., a fault-confirmation signal **114**) generated by the EPGS **105**. In an example with reference to FIG. 2, the fault event may occur and the EPGS **105** may generate the fault-confirmation signal **114** at time point **234** due to a fault condition at time point **233** and/or one or more conditions ranging from time point **233** to time point **234**.

[0052] In some embodiments, the pre-fault snapshot data **140-a** includes a first portion of the stored status signals **112** corresponding to a temporal period **232** (also referred to herein as a fault time delay) prior to a time point **234** associated with occurrence of the fault event. For example, the first portion of the stored status signals **112** may include status signals **112** collected during the

temporal period **232** ranging from time point **233** to time point **234**.

[0053] In some embodiments, the fault snapshot data **140-b** includes a second portion of the stored status signals **112** corresponding to the time point **234** associated with occurrence of the fault event. For example, the first portion of the stored status signals **112** may include status signals **112** collected at the time point **234** or at about the time point **234** (e.g., +/- a temporal amount with respect to the time point **234**).

[0054] In some embodiments, the computing devices **103** (computing device **103-a**, computing device **103-b**) are configured to generate fault analysis data **145** associated with the fault event. For example, the computing devices **103** are configured to generate the fault analysis data **145** based on processing the pre-fault snapshot data **140-a** and the fault snapshot data **140-b**. Aspects of the present disclosure support generating the fault analysis data **145** autonomously and/or semi-autonomously.

[0055] In some embodiments, the aircraft system **102** may transmit the snapshot data record **135** (e.g., pre-fault snapshot data **140-a** and fault snapshot data **140-b**) to the computing device **103-b**. In a non-limiting example, the snapshot data record **135** may include an indication of an overvoltage condition due to one or more factors.

[0056] In some other embodiments, the system **100** supports transferring (e.g., via data transmission) the snapshot data record **135** from the aircraft system **102** to the computing device **103-b** via electrical coupling to the NVM device **125** or electrical coupling to another suitable removable memory storage device on which the snapshot data record **135** may be stored. For example, the system **100** supports data dumps of raw unfiltered data from the aircraft system **102** to the computing device **103-b**, based on which the computing device **103-b** may generate the fault analysis data **145**.

[0057] In some examples, the fault analysis data **145** includes a rate of change associated with at least one status signal **112** of the status signals **112**. In some examples, the fault analysis data **145** includes a pattern associated with at least one status signal **112** of the status signals **112**. For example, the fault analysis data **145** may include a rate of change or a pattern associated with a voltage, a current, a frequency, a temperature, an internal controller status, a state of a switch, or the like of one or more of the LRUs **120**.

[0058] In some embodiments, the fault analysis data **145** may include a correlation between one or more conditions or events (e.g., as determined by a computing device **103** from associated status signals **112**) included in the temporal period **232** to the detected fault event. For example, the fault analysis data **145** may include insight into the status of a given component (e.g., LRU **120**) of the aircraft system **102** leading up to a fault confirmation signal **114**. Accordingly, for example, the techniques described herein provide more effective and more accurate insight of the cause of a fault event compared to other techniques which fail to incorporate pre-fault snapshot data **140-a** into fault analysis.

[0059] In some examples, the fault analysis data **145** includes a temporal indication of the occurrence of the fault condition and the occurrence of the fault event. For example, the temporal indication may include the time point **233**, the time point **234**, and/or a difference value between the time point **233**.

[0060] In some examples, the fault analysis data **145** includes an indication of at least one second fault event included in the pre-fault snapshot data **140-a** or the fault snapshot data **140-b**, wherein the at least one second fault event is of a different type compared to the fault event. In some examples, the fault analysis data **145** includes an indication of at least one second fault condition included in the pre-fault snapshot data **140-a** or the fault snapshot data **140-b**, wherein the at least one second fault condition is of a different type compared to the fault condition.

[0061] In some embodiments, the system **100** is configured to set or tune the temporal period **232** based on a type of the fault event. For example, the computing device **103** (e.g., computing device **103-a**, computing device **103-b**) may determine the type of the fault event based on a

corresponding fault code (e.g., fault-confirmation signal **114**) generated by the EPGS **105**. Based on the type of the fault event, the computing device **103** may set the temporal period **232** for the pre-fault snapshot data **140-a** and assess, in the pre-fault snapshot data **140-a**, status signals **112** included in the temporal period **232**. Accordingly, for example, aspects of the present disclosure support configuring the temporal period **232** for assessing or determining a target type of fault. [0062] Additionally, or alternatively, the computing device **103** may assess, in the snapshot data record **135**, all status signals **112** included in temporal period **231**. Based on the type of the fault event, the computing device **103** may process and analyze, from the temporal period **231**, status signals **112** included in the temporal period **232** and status signals **112** outside the temporal period **232**. It is to be understood that in the examples described herein, the temporal period **232** may include time point **233** and time point **234**, or alternatively, may be separate from time point **234**. [0063] FIG. **3** illustrates an example flowchart of a method **300** that supports enhanced troubleshooting of aircraft EPGS fault events in accordance with one or more embodiments of the present disclosure. The method **300** may be implemented by aircraft system **102** and/or a computing device **103** described with reference to FIG. **1**.

[0064] At **305**, the method **300** includes identifying faults to be detected and stored/reported.

[0065] At **310**, the method **300** includes determining whether to define pre-trigger snapshot data (e.g., pre-fault snapshot data **140-a** described herein). For example, at **310**, the method **300** may include determining whether pre-trigger snapshot data is beneficial for fault analysis. For ‘Yes’, the method **300** may include proceeding to **315**. For ‘No,’ the method **300** may include proceeding to **325**.

[0066] At **315**, the method **300** may include defining a start time for the pre-trigger snapshot data (e.g., a start time point for temporal period **232** described herein).

[0067] At **320**, the method **300** may include defining pre-trigger snapshot data elements (e.g., status signals **112** and associated components of the aircraft system **102** (for example, LRUs **120**) to include in pre-fault snapshot data **140-a** and/or fault snapshot data **140-b**). In some aspects, at **320**, the method **300** may include storing the pre-trigger snapshot data elements (e.g., pre-fault snapshot data **140-a**) to a data record (e.g., snapshot data record **135**) as described herein.

[0068] At **325**, the method **300** may include defining snapshot data (e.g., fault snapshot data **140-b**) for fault at confirmation time. In some aspects, at **325**, the method **300** may include storing the snapshot data to a data record (e.g., snapshot data record **135**) as described herein.

[0069] At **330**, the method **300** may include generating fault reporting and snapshot configuration data (e.g., fault analysis data **145** described herein).

[0070] FIG. **4** illustrates an example flowchart of a method **400** that supports enhanced troubleshooting of aircraft EPGS fault events in accordance with one or more embodiments of the present disclosure. The method **400** may be implemented by aircraft system **102** and/or a computing device **103** described with reference to FIG. **1**.

[0071] At **405**, the method **400** includes continuously capturing, by one or more processors, status signals associated with an electric power generating system of an aircraft and storing the status signals to a circular buffer.

[0072] At **410**, the method **400** includes, in response to detecting a fault event associated with the electric power generating system, storing pre-fault snapshot data and fault snapshot data associated with the fault event to a memory device.

[0073] In some aspects, detecting the occurrence of the fault event is based on receiving a fault-confirmation signal.

[0074] In some aspects, the pre-fault snapshot data includes a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event. In some aspects, the fault snapshot data includes a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

[0075] In some aspects, the method **400** includes setting the temporal period based on a type of the

fault event.

[0076] At **415**, the method **400** includes generating fault analysis data associated with the fault event based on processing, by a computing device, the pre-fault snapshot data and the fault snapshot data.

[0077] In some aspects, the fault analysis data includes a rate of change associated with at least one status signal of the status signals.

[0078] In some aspects, the fault analysis data includes a pattern associated with at least one status signal of the status signals.

[0079] In some aspects, the fault analysis data includes a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.

[0080] In some aspects, the fault analysis data includes an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, where the at least one second fault event is of a different type compared to the fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, where the at least one second fault condition is of a different type compared to the fault condition.

[0081] In the descriptions of the flowcharts herein, the operations may be performed in a different order than the order shown, or the operations may be performed in different orders or at different times. Certain operations may also be left out of the flowcharts, one or more operations may be repeated, or other operations may be added to the flowcharts.

[0082] Set forth are embodiments supported by the present disclosure.

[0083] Embodiment 1. A fault analysis system, comprising: a memory device comprising a circular buffer; and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer; wherein in the one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device, wherein: the pre-fault snapshot data comprises a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data comprises a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

[0084] Embodiment 2. The fault analysis system of Embodiment 1, further comprising: one or more second processors configured to generate fault analysis data associated with the fault event based on processing, by the one or more second processors, the pre-fault snapshot data and the fault snapshot data.

[0085] Embodiment 3. The fault analysis system of any of Embodiments 1 or 2, wherein the fault analysis data comprises a rate of change associated with at least one status signal of the status signals.

[0086] Embodiment 4. The fault analysis system of any of Embodiments 1 through 3, wherein the fault analysis data comprises a pattern associated with at least one status signal of the status signals.

[0087] Embodiment 5. The fault analysis system of any of Embodiments 1 through 4, wherein the fault analysis data comprises a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.

[0088] Embodiment 6. The fault analysis system of any of Embodiments 1 through 5, wherein the fault analysis data comprises an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault event is of a different type compared to the fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault condition is of a different type compared to the fault condition.

[0089] Embodiment 7. The fault analysis system of any of Embodiments 1 through 6, wherein the fault analysis system is configured to set the temporal period based on a type of the fault event.

[0090] Embodiment 8. The fault analysis system of any of Embodiments 1 through 7, wherein the fault analysis system is configured to detect the occurrence of the fault event based on receiving a fault-confirmation signal.

[0091] Embodiment 9. A fault reporting system, comprising: a memory device comprising a circular buffer; and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer; wherein in the one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device, wherein: the pre-fault snapshot data comprises a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data comprises a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

[0092] Embodiment 10. The fault reporting system of Embodiment 9, wherein the fault reporting system is configured to provide the pre-fault snapshot data and the fault snapshot data in association with generating fault analysis data associated with the fault event.

[0093] Embodiment 11. The fault reporting system of any of Embodiments 9 or 10, wherein the fault reporting system is configured to set the temporal period based on a type of the fault event.

[0094] Embodiment 12. The fault reporting system of any of Embodiments 9 through 11, wherein the fault reporting system is configured to detect the occurrence of the fault event based on receiving a fault-confirmation signal.

[0095] Embodiment 13. A method comprising: continuously capturing, by one or more processors, status signals associated with an electric power generating system of an aircraft and storing the status signals to a circular buffer; and in response to detecting a fault event associated with the electric power generating system, storing pre-fault snapshot data and fault snapshot data associated with the fault event to a memory device, wherein: the pre-fault snapshot data comprises a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data comprises a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

[0096] Embodiment 14. The method of Embodiment 13, further comprising: generating fault analysis data associated with the fault event based on processing, by a computing device, the pre-fault snapshot data and the fault snapshot data.

[0097] Embodiment 15. The method of any of Embodiments 13 or 14, wherein the fault analysis data comprises a rate of change associated with at least one status signal of the status signals.

[0098] Embodiment 16. The method of any of Embodiments 13 through 15, wherein the fault analysis data comprises a pattern associated with at least one status signal of the status signals.

[0099] Embodiment 17. The method of any of Embodiments 13 through 16, wherein the fault analysis data comprises a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.

[0100] Embodiment 18. The method of any of Embodiments 13 through 17, wherein the fault analysis data comprises an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault event is of a different type compared to the fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault condition is of a different type compared to the fault condition.

[0101] Embodiment 19. The method of any of Embodiments 13 through 18, further comprising setting the temporal period based on a type of the fault event.

[0102] Embodiment 20. The method of any of Embodiments 13 through 19, wherein detecting the occurrence of the fault event is based on receiving a fault-confirmation signal.

[0103] Any aspect in combination with any one or more other aspects.
[0104] Any one or more of the features disclosed herein.
[0105] Any one or more of the features as substantially disclosed herein.
[0106] Any one or more of the features as substantially disclosed herein in combination with any one or more other features as substantially disclosed herein.
[0107] Any one of the aspects/features/implementations in combination with any one or more other aspects/features/implementations.
[0108] Use of any one or more of the aspects or features as disclosed herein.
[0109] The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.
[0110] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.
[0111] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A fault analysis system, comprising: a memory device comprising a circular buffer; and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer; wherein in the one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device, wherein: the pre-fault snapshot data comprises a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data comprises a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.
2. The fault analysis system of claim 1, further comprising: one or more second processors configured to generate fault analysis data associated with the fault event based on processing, by the one or more second processors, the pre-fault snapshot data and the fault snapshot data.
3. The fault analysis system of claim 2, wherein the fault analysis data comprises a rate of change associated with at least one status signal of the status signals.
4. The fault analysis system of claim 2, wherein the fault analysis data comprises a pattern associated with at least one status signal of the status signals.
5. The fault analysis system of claim 2, wherein the fault analysis data comprises a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.
6. The fault analysis system of claim 2, wherein the fault analysis data comprises an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault event is of a different type compared to the

fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault condition is of a different type compared to the fault condition.

7. The fault analysis system of claim 1, wherein the fault analysis system is configured to set the temporal period based on a type of the fault event.

8. The fault analysis system of claim 1, wherein the fault analysis system is configured to detect the occurrence of the fault event based on receiving a fault-confirmation signal.

2. A fault reporting system, comprising: a memory device comprising a circular buffer; and one or more processors configured to continuously capture status signals associated with an electric power generating system of an aircraft and store the status signals to the circular buffer; wherein in the one or more processors are configured to, in response to detecting a fault event associated with the electric power generating system, store pre-fault snapshot data and fault snapshot data associated with the fault event to the memory device, wherein: the pre-fault snapshot data comprises a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data comprises a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

10. The fault reporting system of claim 9, wherein the fault reporting system is configured to provide the pre-fault snapshot data and the fault snapshot data in association with generating fault analysis data associated with the fault event.

11. The fault reporting system of claim 9, wherein the fault reporting system is configured to set the temporal period based on a type of the fault event.

12. The fault reporting system of claim 9, wherein the fault reporting system is configured to detect the occurrence of the fault event based on receiving a fault-confirmation signal.

3. A method comprising: continuously capturing, by one or more processors, status signals associated with an electric power generating system of an aircraft and storing the status signals to a circular buffer; and in response to detecting a fault event associated with the electric power generating system, storing pre-fault snapshot data and fault snapshot data associated with the fault event to a memory device, wherein: the pre-fault snapshot data comprises a first portion of the stored status signals corresponding to a temporal period prior to a time point associated with occurrence of the fault event; and the fault snapshot data comprises a second portion of the stored status signals corresponding to the time point associated with occurrence of the fault event.

14. The method of claim 13, further comprising: generating fault analysis data associated with the fault event based on processing, by a computing device, the pre-fault snapshot data and the fault snapshot data.

15. The method of claim 14, wherein the fault analysis data comprises a rate of change associated with at least one status signal of the status signals.

16. The method of claim 14, wherein the fault analysis data comprises a pattern associated with at least one status signal of the status signals.

17. The method of claim 14, wherein the fault analysis data comprises a temporal indication of the occurrence of a fault condition and the occurrence of the fault event.

18. The method of claim 14, wherein the fault analysis data comprises an indication of one or more of: at least one second fault event included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault event is of a different type compared to the fault event; and at least one second fault condition included in the pre-fault snapshot data or the fault snapshot data, wherein the at least one second fault condition is of a different type compared to the fault condition.

19. The method of claim 14, further comprising setting the temporal period based on a type of the fault event.

20. The method of claim 14, wherein detecting the occurrence of the fault event is based on receiving a fault-confirmation signal.
